

Aamir Yousuf

Ph.D. Candidate, Computational Mathematics
IIT Bombay and Monash University

Curriculum Vitae
September 2026

Current address

Department of Mathematics
Indian Institute of Technology Bombay
Powai, Maharashtra 400076, India

Mobile: +91 7006587657

Email: aamiryousuf@monash.edu



Research interests

Numerical methods for partial differential equations; finite element, finite difference, and hybrid high-order discretizations; coupled multiphysics and nonlinear plate problems; and stability and convergence analysis of time-dependent problems.

Experience

- Senior Research Fellow, ANRF – J. C. Bose National Fellowship Project. March 2026–Present
Principal Investigator: Prof. Neela Nataraj, Indian Institute of Technology Bombay (IIT Bombay).
Project: *Numerical Analysis of Thermoelastic Diffusion in Thin and Very Thin Plates and Deep Neural-Network Solvers.*

Education

- Ph.D. in Computational Mathematics, Joint Program of IIT Bombay (India) & Monash University (Australia) through IITB-Monash Research Academy. (Thesis submitted) 2022–Present
- Master of Science in Mathematics, University of Kashmir, Srinagar, India. 2017–2020
- Bachelor of Science, General Science, University of Kashmir, Srinagar, India. 2014–2017

Doctoral research

New Discretizations for Coupled Multiphysics Models in Slender Geometries (Ph.D. thesis).

Supervisors: Prof. Neela Nataraj (IIT Bombay) and Prof. Ricardo Ruiz-Baier (Monash University). January 2022–Present

- Developed fully discrete schemes for coupled multiphysics and nonlinear dynamic plate problems, all modeling thin and very thin elastic plates.
- Employed lowest-order nonstandard finite elements for the spatial discretization of the fourth-order equations, conforming finite elements for the second-order equations, and finite-difference schemes for the temporal discretization of all coupled systems.
- Focused on three connected questions: stable and accurate spatial discretization, consistent and efficient discretization of coupled terms, and reliable time integration across equations with different time-stepping schemes.
- Proved well-posedness, stability, and optimal-order spatial and quadratic temporal convergence for all schemes, and designed energy-conserving, unconditionally stable schemes for nonlinear plate problems.

Academic projects

- Fully discrete scheme for the biharmonic wave equation.** July–October 2023
A course project for MA 855 (Topics in Numerical Analysis I) with instructor: Prof. Neela Nataraj, IIT Bombay.
Developed a fully discrete numerical scheme for the biharmonic wave equation and carried out its analysis.
- Finite element discretization of the Euler–Bernoulli beam theory.** January–May 2022
A course project for CE 620 (Finite Element Methods) with instructor: Prof. Mandar Inamdar, IIT Bombay.
Developed the finite element discretization for the Euler–Bernoulli beam theory, jointly with two other project members.

Publications

Journal articles

- N. Nataraj, R. Ruiz-Baier, A. Yousuf, “Unified numerical analysis for thermoelastic diffusion and thermo-poroelasticity of thin plates,” *Mathematical Models and Methods in Applied Sciences*, 2026. [PDF](#)
- N. Nataraj, R. Ruiz-Baier, A. Yousuf, “Semi- and fully-discrete analysis of lowest-order nonstandard finite element methods for the biharmonic wave problem,” *Computational Methods in Applied Mathematics*, 25(4), 921–948, 2025. [PDF](#)

Preprints

- C. Carstensen, N. Nataraj, R. Ruiz-Baier, A. Yousuf, “Unconditionally stable and energy-conserving discretization of the dynamic von Kármán equations,” *arXiv:2608.14087*, 2026. [PDF](#)

4. R. Kumar, N. Nataraj, A. Yousuf, “Hybrid-high order method in space and implicit schemes in time for the biharmonic wave equation,” *arXiv:2606.15163*, 2026. [PDF](#)

Honors and awards

Fellowships and scholarships

- Senior Research Fellow, ANRF – J. C. Bose National Fellowship Project, IIT Bombay. March 2026–Present
- IITB-Monash Research Scholar, Collaborative Ph.D. Program. January 2022–Present
- DBT Fellow, Department of Biotechnology, Government of India. January 2022–Present
- Qualified CSIR-NET with All India Rank 33. 2022
- Qualified GATE with All India Rank 157 and All India Rank 364. 2021 & 2022

Awards and recognition

- Student Member, Australian Mathematical Society. January 2022–Present
- Conference Program Compiler, CTAC 2024, Monash University, Melbourne, Australia. 2024
- Sports Person of the Year, Hostel-13, IIT Bombay. 2025
- Sports Citation of the Year, Hostel-13, IIT Bombay (highest inter-hostel sports honor). 2023
- Member, Inter-IIT Cricket Team, IIT Bombay; participated in national-level tournaments. 2022–2025

Teaching experience (Teaching assistant)

Course	Institution	Period
SI 507 – Numerical Analysis	IIT Bombay	July–October 2025
MTH1010 – Functions and their Applications	Monash University	March–June 2025
MA 214 – Numerical Analysis	IIT Bombay	January–May 2024
MA 111 – Calculus II	IIT Bombay	July–October 2023
SI 416 – Optimization	IIT Bombay	February–May 2023
MA 105 – Calculus I	IIT Bombay	January–February 2023

Conferences and invited talks

- Scientific Computing and Differential Equations (SciCADE-26), Edinburgh, Scotland, UK.
Funded by the Institution of Eminence (IoE), IIT Bombay. 29 June–3 July 2026
- Advanced Computational Methods in Engineering and Applied Mathematics (ACOMEN-25), Ghent, Belgium.
Funded by IIT Bombay. 15–19 September 2025
- Australia and New Zealand Industrial and Applied Mathematics (ANZIAM-25), Coffs Harbour, Australia.
Funded by IITB-Monash student funding, Monash University. 02–05 February 2025
- Joint Meeting of the NZMS, AustMS and AMS 2024, University of Auckland, New Zealand.
Jointly funded by Monash University and AustMS Student Support Scheme. 09–13 December 2024
- Computational Techniques and Applications Conference (CTAC-24), Monash University, Melbourne, Australia.
Funded by the School of Mathematics, Monash University. 19–22 November 2024
- Latest Advances in Computational and Applied Mathematics (LACAM-22), IISER Thiruvananthapuram, India.
Funded by IITB-Monash Research Academy. 21–24 February 2022

Schools and short courses

- Discrete de Rham Method, Prof. Jérôme Droniou (University of Montpellier).
Venue: IIT Bombay, India. February 2026
- The Mathematics of Liquid Crystals, Prof. Apala Majumdar (University of Manchester).
Venue: IIT Bombay, India. February 2026
- Optimality of Adaptive Finite Element Methods, C. Carstensen, L. Diening, C. Kreuzer, D. Praetorius.
Venue: Bonn University, Germany. September 2025
- Adaptive Finite Element Methods, Prof. Carsten Carstensen (Humboldt University Berlin).
Venue: Monash University, Australia. December 2024
- Introduction to HHO Methods, Prof. Jérôme Droniou and Prof. Daniele Di Pietro (University of Montpellier).
Mode: Online. April 2024

Technical skills

- **Programming languages:** C++, Python.
- **Numerical software:** FEniCS, FreeFEM, MATLAB, Firedrake.
- **Tools:** L^AT_EX, Excel, MS Office.

Additional information

- **Strengths:** Mentoring, adaptability, inquisitiveness.
- **Languages known:** English, Urdu, Kashmiri, Hindi, Arabic (beginner).
- **Interests:** Playing and watching cricket, traveling, trekking.

Ph.D. Coursework

Code	Course Title	Credits
MA 855	Topics in Numerical Analysis I	6
MA 854	Topics in Differential Equations II	6
MA 824	Functional Analysis	6
MA 817	Partial Differential Equations I	6
MA 814	Complex Analysis	6
MA 813	Measure Theory	6
AE 738	Tensors for Engineers	6
CE 620	Finite Element Methods	6
MA 540	Numerical Methods for Partial Differential Equations	6
MA 503	Functional Analysis	8
MES 801	Seminar	4
ME 899	Communication Skills	6
TA 101	Teaching Assistant Skill Enhancement & Training (TASET)	–
GC 101	Gender in the Workplace	–

References

1. Prof. Neela Nataraj

Professor, Department of Mathematics
Indian Institute of Technology Bombay
Powai, Mumbai 400076, India

Phone: +91-22-2576 7468

Fax: +91-22-2572 4450

Email: neela@math.iitb.ac.in

2. Prof. Ricardo Ruiz Baier

Professor of Computational Mathematics
School of Mathematics, Monash University
Office 315, 9 Rainforest Walk
Melbourne, VIC 3800, Australia

Phone: +61 3 9905 4485

Email: ricardo.ruizbaier@monash.edu

3. Prof. Harsha Hutridurga

Associate Professor, Department of Mathematics
Indian Institute of Technology Bombay
Powai, Mumbai 400076, India

Phone: +91-22-2576 9474

Email: hutridurga@iitb.ac.in